

Where's the Water? Supporting Clean Water Access for the Homeless Community

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ABSTRACT

Access to clean water is essential, yet it poses a significant challenge for the homeless population. Our project, 'Where's the Water,' is a web-based tool designed to improve water access for the homeless community. It maps nearby clean water sources like drinking fountains, public restrooms, and showers. The tool's design was informed through interviews with the homeless community in Ann Arbor, Michigan. The insights gained from these interviews were further supported by key findings from recent studies related to homelessness and water access. Besides locating, our tool's functionality also includes filtering sources for operational hours and water quality. It features crowd-sourcing, allowing users to add new sources on the map, effectively utilizing community knowledge. In this article, we describe our research and design approach, highlighting the community and organizational feedback that helped turn our concept into a useful tool.

CCS CONCEPTS

• Human-centered computing → User centered design.

KEYWORDS

Clean Water Access, Homeless Population, Health, User-centered Design

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1 INTRODUCTION

For many people, especially those experiencing homelessness, reliable and consistent access to clean water is out of reach. According to survey research [12] among the homeless population in San Francisco, 50 out of 73 respondents reported facing daily obstacles in fulfilling their fundamental water needs. This is a challenge faced

globally. In studies carried out among the street population in Belo Horizonte, Brazil [8], access to water and bathrooms is viewed more as a charitable provision than a basic human right. The water access needs of homeless communities worldwide are still largely unmet, a concern that is acknowledged in the UN Sustainable Development Goals, specifically under the clean water and sanitation target [7].

The challenges in water access for the homeless stem from two main issues: scarcity and unaffordability of public water sources, and a lack of information about available resources. People in the area facing water scarcity have to limit their water usage and travel long distances to access water because there are very few public water sources available nearby [12, 16]. Some government and non-profit organizations are actively taking action to combat scarcity in their outreach programs, such as introducing a mobile hygiene trailer offering free showers and laundry services [11] and distributing drinking water to the unsheltered community [1].

In addition to water scarcity, a lack of information about resources exacerbates the difficulty of water searching. Homeless individuals often spend hours looking for water sources, only to find them closed or unavailable, which prolongs their search. In Phoenix, Arizona, more than 15% of public water resources, such as fountains and restrooms in parks, are non-functional [4]. Some water sources have hours of operation and an unfixed schedule which adds more difficulty: access to public drinking fountains is often restricted due to park closures [12], many businesses that might offer water donations close at night [16], and water trucks do not operate on a fixed schedule [4]. This information is not always readily available or transparent, making it difficult for homeless people to plan their water collection.

There are currently few solutions aimed at addressing the information gap related to locating clean water sources. Our conversations with individuals experiencing homelessness revealed that existing solutions do not provide the comprehensive information required by the community. Our platform is designed to bridge this gap by helping the homeless community in finding nearby water sources, offering clear details about the types of sources, their hours of operation, and water conditions - information highly valued by the community. Additionally, in line with the community's tradition of community-driven information sharing, our platform features crowd-sourcing capabilities that allow users to add new water sources.

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2 BACKGROUND AND RELATED WORK

The rising trend of homelessness in the United States is evident both nationally and regionally [2]. A significant portion of those experiencing homelessness live in unsheltered conditions; in California, for example, the number reached 76% [10]. Without stable housing, individuals are forced to rely heavily on social infrastructure and interpersonal networks for water access, leading to considerable inaccessibility and inconsistency [4, 6].

Key obstacles include being denied access to water in private and public spaces, such as restaurants and libraries, often because of appearance or the requirement to purchase something [12]. Another issue is the limited availability of resources, including public restrooms in parks and libraries, and drinking water from shelters, organizations, and businesses, which are unavailable at night [12, 16]. In the Project for Sanitation Justice study, a participant reported that due to the lack of overnight restroom access, they stopped eating and drinking after 4 pm [3]. Some public water sources are even non-functional [4]. Therefore, the effort to access these essential resources often involves a significant investment of time and energy, with individuals having to learn through trial and error [6].

To tackle the problem of limited water access, innovative solutions like Data SF's Map of Public Water Fountains [15] and the Project for Sanitation Justice's Bathroom Mapping [17] are now available online. These tools are designed to assist individuals who depend on public water sources, enabling them to easily locate nearby options. The Bathroom Mapping goes a step further by offering comprehensive information such as operational hours, facilities like baby changing stations, and more, allowing users to choose the most appropriate restroom based on their needs.

However, there's a need to better tailor such informational resources to the needs of the homeless community. This involves broadening the network to include diverse water types and providing real-time guidance on locating water sources, along with immediate updates on their availability and conditions. While the earlier mentioned Map of Public Water Fountains and Bathroom Mapping tool currently focuses on public sources, incorporating additional sources such as friendly bars, restaurants, and gas stations - often seen as major water donors [8] - is crucial. Streamlining the search process to display available sources at any given moment can save time and energy. Moreover, the capability to swiftly update information about water source availability or conditions is essential. Improving these aspects would greatly increase the utility of such tools for populations like the homeless, who are heavily dependent on these resources.

3 USER RESEARCH AND DESIGN APPROACH

Before holding interviews with members of the homeless population, we consulted experts to ensure our research was conducted respectfully and in line with best practices for engaging with vulnerable populations. Our selection criteria focused on their experience with such groups and their availability within our project timeline. Ultimately, we spoke with five individuals, including two non-profit organization workers, two university professors, and an employee

from the Centers for Disease Control and Prevention (CDC). Following our discussions with experts, we revised our interview protocol to incorporate their advice.

Following our consultations with experts, we conducted qualitative research with the street population in Ann Arbor, Michigan, in October 2023. Our research focused on understanding their daily routines, methods of accessing water, experiences in obtaining water, and concerns about water accessibility. We introduced our study at a local free meal event, where we explained our goals and methodology. Four individuals experiencing homelessness showed interest in participating. We conducted individual interviews with them, offering a \$30 Visa gift card as compensation. We emphasized their right to withdraw from the study at any time. The interviews were semi-structured, based on each researcher's experience, and informed by feedback and suggestions. This study was reviewed and received an exemption from the Institutional Review Board (IRB) at the University of Michigan.

The interview audios were then synthesized and analyzed using thematic coding and affinity diagramming. Relevant first-level themes were extracted through this process and were subsequently refined into several key findings. After deriving insights from our research, we established design requirements tailored to the homeless community's needs. Our iterative design process began with a low-fidelity prototype tested by three University of Michigan students to pinpoint basic usability issues, followed by a high-fidelity prototype evaluated by eleven homeless individuals across two rounds of testing to verify if our product met their needs. These participants were recruited from various free food events and were compensated with \$30 in cash for their participation. In all tests, participants were given specific tasks to perform using the prototype, followed by post-test questions to collect feedback on their user experience. Based on the feedback gathered, iterations and refinements were made to the prototype.

4 RESEARCH FINDINGS

The following sections describe our findings from expert consultations and interviews with members of the homeless population, as well as how these findings informed our design.

4.1 Expert Consultations

Experts advised community-based participatory research (CBPR) approach, promoting collaboration and equal knowledge contribution between community members and researchers [9]. Following expert advice like "always follow the lead of the person and practice active listening", we ensured a non-judgmental environment by avoiding assumptions in our questioning in subsequent interviews. For instance, we neutrally asked about any problems with water access rather than presuming it to be a challenge, allowing for an open response. If issues were acknowledged, we probed further for details. Participants had enough time to reflect and share their experiences.

Experts highlighted the importance of using straightforward and accessible language to foster effective bi-directional communication. As one expert told us, "Meet participants where they are." Recognizing the diverse backgrounds within the homeless community,

we reviewed our interview transcripts to eliminate academic jargon and employ terminology more universally understood across various demographics. For instance, we opted to consistently use the term "homelessness" instead of "instability," which was initially used in our questions regarding their experiences. This approach aimed to ensure clarity and inclusiveness in our dialogue.

Experts highlighted the importance of interviewing in safe, comfortable locations and recommended food giveaways as ideal venues. One expert suggested we "Bring coffee and snacks. Sit down to talk with them." Following their advice, we connected with participants at such events, creating a relaxed atmosphere. It's also essential to transparently communicate the purpose and details of the research, including the benefits, risks, and alternatives. Another expert that we spoke with emphasized, "Be realistic. Explicitly tell them what you can do and what you cannot do." Thus, at the start of our study, we clearly outlined our project's goals and what participants could anticipate during the interviews, ensuring they were fully informed and comfortable proceeding.

Offering suitable incentives is important to encourage participation from the community and to demonstrate respect for their time. We were advised to review other studies to gauge standard practices and to also consult with our partnering organizations. We finally chose to provide \$30 and a basic reusable water bottle. This approach aimed to adequately meet the participants' needs and align with the theme of our project.

4.2 Interviews With the Homeless Population

4.2.1 Knowing Where to Go. To navigate around uncooperative businesses, individuals typically employ trial and error or rely on word of mouth. They approach businesses directly to inquire about water or restroom access, moving on if denied. Over time, this method allows them to build a mental map of welcoming locations. One participant told us, "[I know where to access clean water through] experience and word of mouth. Avoid certain businesses" (P3). However, accessing water at night becomes particularly challenging as businesses and shelters close. We had somebody tell us that "When it reaches 10-11 o'clock and most restaurants close, it becomes more of a challenge for accessing water" (P4). Additionally, public water fountains often become inaccessible, being situated behind gates of parks that are closed during the late hours [12]. Newcomers to the community often benefit from the shared knowledge of more experienced individuals.

The trial-and-error approach is hit or miss and inefficient, potentially leaving individuals dehydrated for extended periods, especially in extreme weather conditions. One individual we interviewed shared a distressing experience. She faced repeated rejections as most establishments dismissed her upon sight. By the time she finally found a place willing to help, she was so severely dehydrated that her body couldn't retain the water she so critically needed. She told us, "I got so dehydrated that by the time I finally got water, I couldn't keep it down" (P3). This story highlights the dire circumstances and urgent need for more compassionate and accessible water provisions for vulnerable populations.

4.2.2 Obtaining Water From Places of Business. Businesses play a crucial role in providing access to drinking water and bathrooms for the homeless, which is also highlighted by several studies [8,

12]. Instead of carrying around water bottles, many individuals prefer to obtain water from select businesses. Participants told us, "In urban places, it is easy to walk in and get free water like in Starbucks and public restaurant bathrooms," and that a lot of the members of the homeless community in Ann Arbor, MI "Always use tap water from Starbucks" (P1). However, this approach faces challenges in certain contexts, like in rural areas where businesses are sparse or during events such as the COVID-19 pandemic when many establishments were closed as precautionary measures. These situations significantly limit the availability and accessibility of such essential resources [12].

Access to water from businesses is not always guaranteed, as evidenced by multiple studies. The homeless community often faces negative experiences, such as being denied access due to their appearance [12], being charged for water [16], or even facing threats to personal safety, as revealed in our interviews. We heard a first-hand account of such difficulty, in which a participant told us, "Recently [my father] got a trespassing citation from a coffee shop. He went in there and he was going to buy a coffee, but he went to use the restroom first... They called the police because they assumed he wasn't going to buy anything" (P3). They want to stay unnoticed, particularly in busy areas, to avoid drawing attention and potentially encountering trouble.

4.2.3 Water Cleanliness. The preference for business-sourced water is partly influenced by the stigma that cheaper or public water is less clean and healthy, as noted in studies [4, 5]. In our interviews, when asked about "clean public water," one participant told us, "I always think of all the bad stuff in water... [We need] more filters to clean through stuff" (P1). Responses such as this reflect a profound distrust in the quality of public water sources. Many are unsure about the safety of available water and its implications for their health. This skepticism towards public water quality is corroborated in broader research; for example, 18.8% of 191 respondents in a more extensive survey expressed apprehension regarding water safety and access [13]. Additionally, some individuals reluctantly consume whatever water is available, despite being cognizant of potential contamination [13].

5 DESIGN REQUIREMENTS

Our interviews yielded critical insights, enabling us to define detailed design requirements that align with our stated project goals, as shown in table 1. This guided us in tailoring our design to effectively meet the specific needs.

6 DESIGN OF WHERE'S THE WATER

6.1 Overview

"Where's the Water?" is a web-based map tool we designed to display nearby water sources. We chose a web-based approach for three reasons: universal access on any internet-enabled device, the frequent and consistent internet use by emerging adult homeless individuals [14], and the ability to print maps from the browser for those without consistent internet access. Homeless individuals can print maps at public libraries, and local organizations or businesses can provide printouts for those in need. Our design includes mobile

Table 1: Design Requirements and Rationale

Requirement	A Description	Rationale
Accessible Information	Users should be able to easily locate nearby water sources, including their type and status. The tool’s design must display buttons with clear labels, ensuring users can fully utilize the tool’s capabilities.	The primary goal expressed by our interviewees is straightforward: to obtain clean water. Our tool should be capable of assisting them, irrespective of their literacy and cultural background.
Nighttime Water Access	Our tool should provide a clear display of the operating hours in the details for each water source. Users can filter these sources based on their hours of operation to view water sources that are currently available.	Interviewees have expressed that locating water sources during late hours is extremely difficult. Therefore, our tool must identify and display water sources that are available at night.
Water Safety Transparency	The tool should include filters for assessing water conditions, allowing users to ensure the safety and cleanliness of the water sources they choose.	Interviewees have expressed concerns about the quality of public water, leading to hesitation in its use. They want to be able to access more detailed information, such as whether the water is safe to drink.
Encourage Public Contributions	Our design should encourage user contributions, creating a community-driven platform where water source information can be shared and updated.	Word of mouth plays a significant role in discovering available water sources. The community relies heavily on the knowledge network as a crucial source of information.

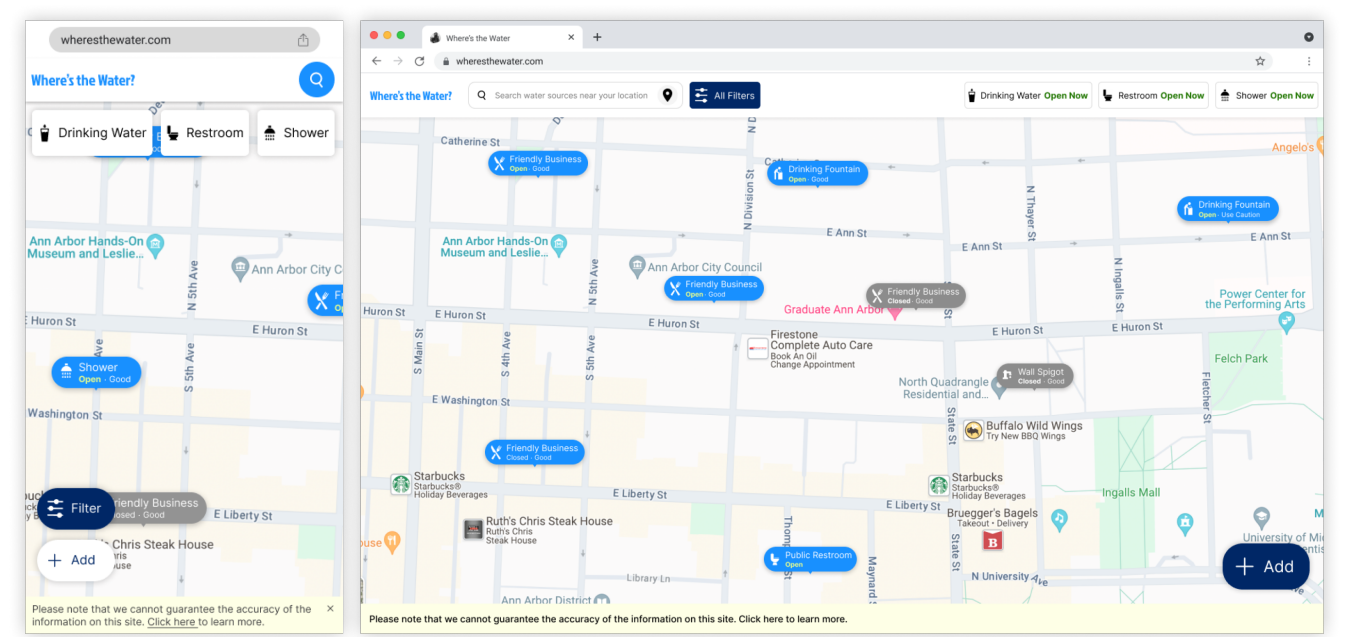


Figure 1: Where’s Water Interface: Positioned at the left is the desktop version, and at the right is the mobile version, displaying water sources information and locations.

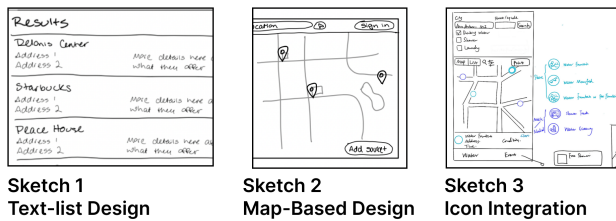


Figure 2: Our Water Source Display evolved through three stages: 1) Text-Based Design, initially using a text-list format; 2) Map-Based Design, with clear location pins; and 3) Icon Integration, adding distinct icons for easy identification of different water sources.

and desktop versions, as shown in Figure 1, ensuring accessibility and functionality on all devices.

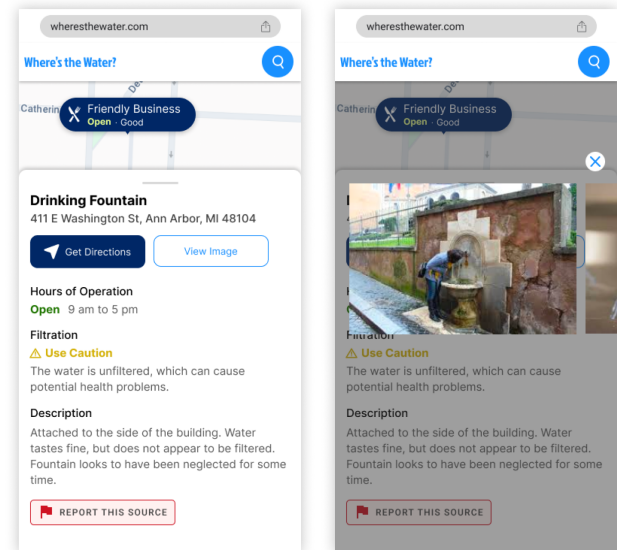
6.2 Key Product Features

6.2.1 Simplified Water Source Display. We designed a user-friendly interface that maps out water sources, making it easier for users to locate drinking water, showers, and restrooms. The map layout was a key feature that we converged upon during the initial ideation phase. We worked separately to sketch design ideas that were discussed during a later team meeting. In the meeting, we decided to focus on designs where the map was the dominant element, believing this approach would make it easier for users to quickly identify and understand the locations of various water sources, as shown in Fig 2.

To enhance clarity on the map, we use distinct icons for each type of water source: for instance, icons for friendly businesses, drinking fountains, showers, and water spigots. We also employ color coding to show availability – with blue indicating an open source and gray signifying a closed one. Additionally, we include text descriptions for each source to ensure accessibility and provide clear information to all users. Clicking a pin expands a panel with details such as location, image, condition, and description, as shown in Fig 3.

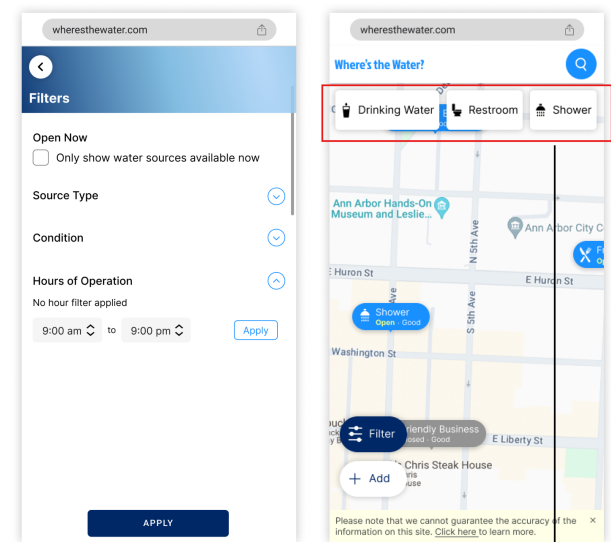
6.2.2 Nighttime Search Functionality. Acknowledging the challenge of finding water sources after hours, we integrated a feature that allows users to filter by hours of operation, enhancing the tool's utility around the clock, as shown in Fig 4. However, our first round of usability testing revealed that users wanted a more efficient way to find water sources that are available when they need them. They wanted a method that eliminates the need for manually inputting operating hours. To accomplish this, we added 'quick filters' at the top of the landing page for easy access. This feature enables users to quickly locate what they need using predefined filters, eliminating the need to navigate through more detailed filter settings.

6.2.3 Water Quality Filters. The tool includes filters for assessing water conditions, allowing users to ensure the safety and cleanliness of the water sources they choose. These filters are accessible via the "conditions" section of the filters panel, as shown in Fig 5. However, we noticed that some participants were having difficulty accessing



Mobile - Water Source Details

Figure 3: Water Source Detail, allowing users to view specifics about each water source and obtain directions.



Mobile: Nighttime Search Functionality

Mobile: Quick Filters for Nighttime Search Functionality

Figure 4: Filters for selecting specific opening times (left). Quick Filters for one-click common requests with just one click/tap (right).

the filters panel during our second round of usability testing. While participants using the desktop version found the filter panel easily accessible, two participants had difficulty locating the buttons to open the filter panel in the mobile version. For this reason, we

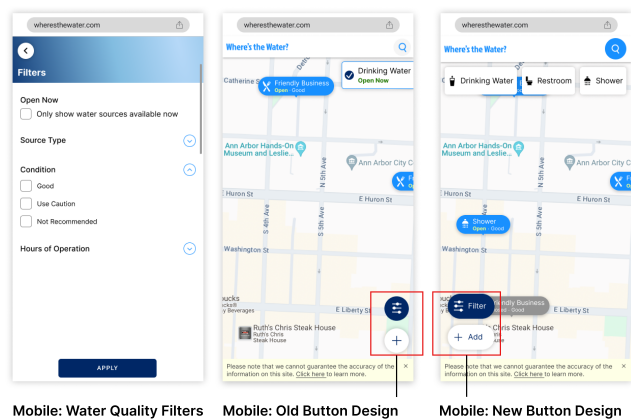


Figure 5: On the Filter page (left), users can now select water sources by quality. The "Filter" and "Add Water Source" buttons are improved from the previous icon-only design (middle) to both icons and descriptions (right).

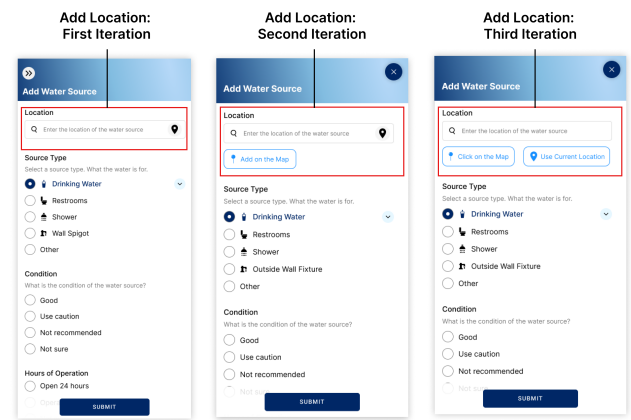


Figure 6: Add Source Panel Iterations (from left to right), where it was made easier for users to be able to click/tap on the map to indicate a source location.

changed the mobile version’s button design to explicitly state its function with text, rather than just relying on an icon.

6.2.4 Community Contributions. Our design encourages user contributions, creating a community-driven platform where water source information can be shared and updated, as shown in Fig 6. In our second round of usability testing, users expressed a desire to be able to click or tap on the map to indicate where a new water source is located. To simplify the process of adding a location, we added a button that would enable users to do this. But we found during our third round of usability testing that users either did not notice the button or confused it with the button that is used to enable location services. Therefore, we listed two buttons with explicit actions in the final version: ‘Click to Add Source’ and ‘Use Current Location’ to show their purpose: click and use.

7 CONCLUSION AND FUTURE WORK

Our team has designed a user-friendly tool to help the homeless community easily locate and report water sources. In-depth interviews with the local homeless population revealed critical unmet needs related to water access. These include knowing where to find water, the ability to locate water sources during nighttime, and understanding the cleanliness of available water. These insights shaped our product’s design requirements, ensuring they align closely with the community’s needs. The design has been continuously refined through iterative testing and direct feedback, particularly from individuals within the homeless community. Test results have shown that users highly value the tool’s map and water filter features, as these enable them to easily find the specific type of water they need and identify currently available sources. The positive reception of these features confirms that our product development is progressing effectively and meeting its intended goals.

Our next goal is to develop and implement this design into a practical tool that will reach as many people as possible through finding funding and developers for this project. We began our focus in Ann Arbor, MI, but our goal is to expand to other U.S. cities and even globally with more local research. Our ultimate objective is to work towards making water more accessible as a fundamental human right.

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